Customer Segmentation & Repeat-Purchase Prediction

Project Objective

Group customers by buying behavior and predict repeat buyers to support targeted marketing.

Data & Feature Engineering

Portfolio reconstruction dataset: 3,900 customer records. RFM-style features were created using recency, purchase frequency and monetary value, with behavioral fields such as previous purchases, purchase frequency, discounts and ratings.

Segmentation Approach

A four-segment customer framework is provided for portfolio demonstration. The original project description specified K-Means with k=4 validated using elbow and silhouette methods; the supplied original raw dataset was not available in the current files, so the attached dataset is a reconstructed portfolio dataset rather than the original source data.

Prediction Approach

A Logistic Regression workflow is documented for repeat-buyer classification. The target is Repeat_Buyer, with behavioral variables used as predictors. Accuracy should be recomputed if the original dataset is restored; the earlier project description reported approximately 80% accuracy.

Business Use Cases

High-value and loyal customers can receive premium offers; regular customers can receive cross-sell campaigns; at-risk customers can receive win-back offers; low-engagement customers can receive onboarding or discount campaigns.

Tools

Python, Pandas, NumPy, Scikit-learn, Matplotlib/Seaborn.

Important Data Note

Because the original raw data was deleted, this package uses a clearly labelled reconstructed dataset for GitHub/portfolio structure. It should not be represented as the original dataset.